

Program: Diploma in Civil Engineering and Planning	
Course Code: 4392	Course Title: Estimation, Tendering and Valuation
Semester: 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To estimate the quantities of item of works involved in Civil Engineering projects.
- To perform rate analysis for different items associated with construction projects.
- To prepare typical tender documents for Civil Engineering work.
- To get acquainted with rent fixation and valuation of Civil structures.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Building Drawing and Planning		Building Drawing and Planning	3
Engineering Mathematics		Engineering Mathematics	1 and 2
Building construction and construction materials		Building construction and construction materials	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Understand the principles and procedure for estimation and costing of Civil Engineering works	14	Understanding
CO2	Prepare detailed estimates for infrastructure projects	17	Applying
CO3	Perform analysis of rates for different items of works and familiarize with the procedure of tendering associated with construction projects	15	Applying
CO4	Assess the value of a property value and fix rent of a building	12	Applying
	Series tests	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	3					
CO3	3					3	
CO4		3			3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive level
CO1	Understand the principles and procedure for estimation and costing of Civil Engineering works		
M1.01	Understand the purpose of estimation and the requirements of an estimator. Explain the rules of measurement as per IS:1200	3	Understanding
M1.02	List and explain different types of estimates.	3	Understanding
M1.03	Understand different stages of preparing Detailed estimate.	4	Understanding
M1.04	Compute volume of earth work by different methods.	4	Applying

Contents:

Estimation - Meaning, purpose, Administrative approval, Technical Sanction and Budget provision.

Types of estimates - Approximate and Detailed estimate. Types and Uses of Estimates: Revised estimate, Supplementary estimate, Repair and maintenance estimate, renovation estimate. Roles and responsibility of Estimator.

Measurement Book, Abstract sheet, Face sheet. Modes of measurement and desired accuracy in measurements for different items of work as per IS:1200. Rules for deduction in different category of work as per IS:1200. Description / specification of items of building work as per PWD /DSR,

Approximate estimate - Definition, Purpose. Methods of approximate estimate - Service unit method, Plinth area rate method, Cubical content method, Typical Bay method, Approximate quantity method (with simple numericals)

Detailed Estimate - Definition and Purpose, Data required for detailed estimate - Civil cost, GST, Contingencies, Supervision charges, Agency charges, Procedure for preparation of detailed estimate. Provisions in detailed estimate-work charged establishment, percentage charges, water supply and sanitary charges and electrification charges etc. Bill of quantities.

Earth work computation – Quantities for roads, embankment and canal by- Mid section area method, mean sectional area method, Prismoidal and Trapezoidal formula.

CO2	Prepare detailed estimates and abstract of estimate for infrastructure projects		
M2.01	Understand different methods of preparing Detailed Estimate.	3	Understanding
M2.02	Prepare detailed estimate and abstract of estimate of a flat roof building.	4	Applying
M2.03	Prepare detailed estimate and abstract of estimate of tiled roof building, water tank and compound wall.	4	Applying
M2.04	Prepare detailed estimate for culvert and retaining wall.	6	Applying
	Series Test I	1	

Contents:

Centre line method and long wall and short wall method

Detailed estimate and abstract of estimate of various items of works for- single storied building with flat roof and one room building with tiled sloping roof (Hip roof), detailed estimate and abstract of estimate of water tank, compound wall, culverts (without splayed wing wall), RCC Retaining walls.

CO3	Perform analysis of rates for different items of works and familiarize with the procedure of tendering associated with construction projects		
M3.01	Understand the importance of rate analysis and familiarize with Standard data book and Schedule of rates.	3	Understanding
M3.02	Prepare data and analysis of rates for various items of work.	5	Applying
M3.03	Familiarize with the process of tendering	4	Understanding
M3.04	Prepare typical tender document for civil engineering work.	3	Understanding

Contents:

Rate Analysis: Definition, Purpose and importance, Lead (Standard and extra), Lift, overhead charges, contractor's profit, water charges, overhead profit, Local Market Rate, Cost Index, Introduction to the use of CPWD data book and schedule of rates, Procedure for rate analysis. Preparation of data and analysis of rates for various items of work connected with building construction and other civil engineering structures

Tender and Tender Documents: Definition of tender, necessity of tender, types of tender Local, Global, Limited. E -Tendering System – Online procedure of submission and opening of bids (Technical and Financial). Notice to invite tender (NIT)- Points to be included while drafting tender notice, Drafting of tender notice. Procedure of submitting filled tender Documents (Two envelope system), procedure of opening tender, comparative statement, scrutiny of tenders, award of contract, letter of award. Meaning of terms - Earnest Money Deposit (EMD), Performance Security Deposit, Validity period, corrigendum to tender notice and its necessity, Unbalanced bid.

Tender documents – Index, tender notice, general instructions, special instructions, Schedule A, Schedule B, Schedule C etc.

CO4	Assess the value of a property value and fix rent of a building		
M4.01	Understand the purpose of Valuation and factors affecting valuation.	3	Understanding
M4.02	Compute Depreciation of properties by different methods	3	Applying
M4.03	Understand the different methods of Valuation	3	Understanding
M4.04	Fix the rent of a building and understand types of lease	3	Applying
	Series Test II	1	
Contents: Valuation – Definition and purpose of Valuation, role of valuer. Definition Cost, Price and Value, Characteristics of value, Factors affecting value. Types of value- Scrap Value, Salvage value, Book Value, Market Value, Distress Value, Sentimental Value, Depreciation, Obsolescence, Sinking fund, Years purchase, Methods of Calculation of Depreciation – Straight line method, Constant Percentage method, Sinking fund method. Valuation of real properties - rental method, profit based method, depreciation method. Valuation of landed properties - belting method, development method, hypothetical building scheme method. Rent fixation. Lease, Types of Lease, Free hold property and Lease hold property.			

Text /Reference:

T/R	BookTitle /Author
T1	Datta, B.N., Estimating and Costing in Civil engineering, UBS Publishers Distributors Pvt. Ltd., New Delhi.
T2	Sharma S C and Deodhar S V, Construction Engineering and Management, Khanna Book Publishing, New Delhi
R2	Rangwala,S.C., Estimating and Costing, Charotar Publishing House PVT. LTD., Anand.
R3	Birdie,G.S., Estimating and Costing, Dhanpat Rai Publishing Company(P) Ltd. New Delhi.
R4	V N Vazirani& S P Chandola, Civil engineering Estimating and Costing, Khanna Publishers
R5	IS 1200-1968; Methods of measurement of Building & Civil Engineering works.
R6	Chakraborti,M., Estimating and costing, specification and valuation in civil engineering, Monojit Chakraborti, Kolkata. 7. PWD Schedule of Rates.
R7	CPWD Schedule of Rates for Delhi-2018
R8	CPWD Analysis of Rates for Delhi-2018, Vol I and II
R9	Ministry of Road Transport and Highways (MORT&H) Specifications and Analysis of Schedule of Rates.
R10	IRC SP: 20 Rural Roads Manual

Online resources:

Sl. No	Website Link
1	https://nptel.ac.in/
2	http://egyankosh.ac.in/
3	https://swayam.gov.in/
4	https://etenders.kerala.gov.in/nicgep/app